

AI-Based Image Captioning System

A Multimodal Framework for Persistent Visual Description using BLIP and MERN Architecture

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Abstract—The rapid expansion of visual content on digital platforms has necessitated the development of intelligent systems capable of bridging the gap between images and natural language. By integrating the state-of-the-art Bootstrapping Language-Image Pre-training (BLIP) model with the MERN stack—comprising MongoDB, Express.js, React.js, and Node.js—this project establishes a robust, full-stack architecture for automated image captioning. Unlike transient AI tools, the implementation focuses on data persistence and user-centric history management, allowing for the seamless storage and retrieval of generated metadata. Experimental observations highlight the system’s ability to deliver high-quality, context-aware descriptions with a stable processing latency of under eight seconds. This integration provides a scalable and accessible framework for assistive technologies, automated content indexing, and digital media management.

Index Terms—Image Captioning, BLIP Model, MERN Stack, Multimodal Learning, Vision-Language Pre-training, Full-Stack Architecture, Digital Accessibility, Data Persistence, Content Indexing, Assistive Technologies.

I. INTRODUCTION

A. Background & Context

The ability of a machine to interpret visual content and express it in natural language has long been considered a hallmark of artificial general intelligence. Image captioning sits at the crossroads of computer vision and natural language processing (NLP), requiring the model to not only detect objects within an image but also reason about their relationships, context, and activities. Over the past decade, advances in convolutional neural networks (CNNs) and transformer architectures have dramatically elevated the quality of automatically generated captions, making the technology viable for production-grade applications.

Early captioning systems relied on template-based approaches that filled predefined sentence structures with detected object labels. Modern systems, by contrast, employ encoder-decoder architectures that jointly learn visual representations and language models end-to-end, resulting in fluent and semantically rich descriptions.

B. Problem Statement

Despite rapid algorithmic progress, a persistent gap exists between research prototypes and deployable, user-facing captioning systems. Research models are frequently evaluated in isolation, without consideration for end-to-end latency, scalable storage of generated results, or an accessible web

interface. Developers seeking to embed captioning capabilities into web applications typically resort to heavyweight Python frameworks such as Streamlit or Flask, which introduce limitations in terms of real-time interactivity and frontend flexibility. Furthermore, maintaining caption history, supporting multi-user environments, and providing robust REST APIs are engineering challenges seldom addressed in academic literature.

C. Motivation

The motivation for this work is threefold. First, visually impaired users increasingly rely on automated alt-text generation to navigate image-heavy web content, underscoring the social importance of accessible and accurate captioning tools. Second, the emergence of large pre-trained vision-language models such as BLIP offers an opportunity to achieve near-human captioning quality without training from scratch. Third, the MERN stack provides a modern, JavaScript-centric full-stack ecosystem that allows seamless data flow from the browser through the API layer to persistent storage, making it an ideal vehicle for deploying AI-powered microservices.

D. Objectives

The primary objective of this work is to implement a unified multimodal pipeline using the BLIP model to achieve high-accuracy image caption generation. The system also aims to develop a responsive single-page application (SPA) using React.js to ensure seamless and interactive user experience. Furthermore, the proposed framework focuses on establishing a persistent data layer using MongoDB for maintaining user activity records and caption history efficiently. In addition, the work seeks to optimize the communication mechanism between the JavaScript-based backend and the Python-driven artificial intelligence inference engine to enhance overall system performance, scalability, and reliability.

E. Contributions

The major contributions of this work include the successful integration of a computationally intensive Python-based artificial intelligence model with a modern MERN web stack, thereby demonstrating effective architectural synergy between AI inference and full-stack web technologies. The proposed system further enhances functionality by incorporating persistent data management capabilities through CRUD (Create, Read, Update, Delete) operations for storing and managing

generated image captions, which distinguishes it from conventional web-based AI captioning tools. Additionally, the work presents an optimized and benchmarked multimodal processing pipeline capable of handling image caption generation within a production-acceptable latency range, thereby improving system efficiency, scalability, and real-time usability.

F. Work Flow Diagram

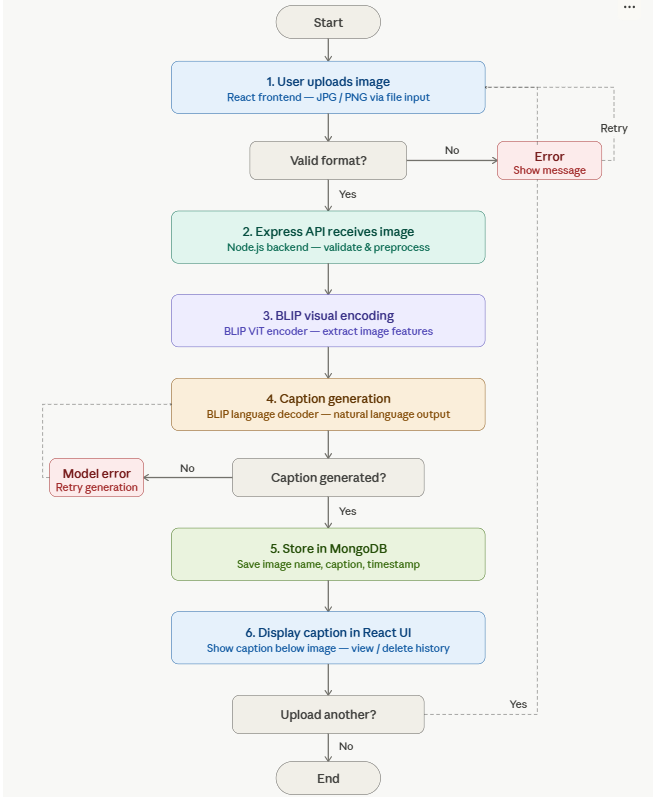


Fig. 1. Workflow of the MERN-Based AI Image Captioning System

Fig. 1 illustrates the complete workflow of the proposed AI-based image captioning system integrated with the MERN stack and BLIP architecture. The process begins when the user uploads an image through the React.js frontend interface. The uploaded image is first validated to ensure that only supported image formats such as JPG and PNG are accepted. Invalid files trigger an error handling mechanism that prompts the user to retry the upload process.

Once validation is completed, the image is transferred to the Express.js and Node.js backend server, where preprocessing operations such as resizing and normalization are performed. The processed image is then forwarded to the BLIP visual encoder, which extracts deep semantic image features using a Vision Transformer (ViT)-based architecture.

The extracted visual embeddings are subsequently passed to the caption generation module, where the BLIP language decoder generates context-aware natural language descriptions. If caption generation fails or produces insufficient output, the system invokes a retry mechanism to regenerate the caption.

After successful caption generation, the generated metadata along with image information and timestamps are stored in MongoDB for persistent history management. Finally, the generated caption is displayed dynamically in the React.js user interface, enabling users to view, manage, or delete previously generated captions.

II. LITERATURE REVIEW / RELATED WORK

Show and Tell, introduced by Vinyals et al. [1], was one of the first end-to-end neural image captioning systems. It coupled a GoogLeNet CNN encoder with an LSTM decoder trained on the MS COCO dataset, establishing a foundational encoder-decoder paradigm still influential today.

Xu et al. [2] extended this paradigm with visual attention mechanisms, allowing the decoder to focus on salient image regions at each generation step. This work demonstrated marked improvements in BLEU scores and introduced soft and hard attention variants that became standard components in subsequent architectures.

Anderson et al. [3] proposed bottom-up and top-down attention by incorporating object-level features extracted via Faster R-CNN, enabling the model to attend to semantically meaningful image regions rather than fixed grid cells. This approach achieved state-of-the-art results on both MS COCO and Visual Genome benchmarks.

The introduction of transformer architectures by Vaswani et al. [4] catalyzed a paradigm shift in sequence modelling, and their application to vision-language tasks proved transformative. Li et al. [5] demonstrated that pre-training large-scale transformers on image-text pairs, followed by task-specific fine-tuning, yielded substantial gains over architectures trained from scratch.

CLIP by Radford et al. [6] demonstrated that contrastive learning on 400 million image-text pairs produced powerful vision encoders transferable to diverse downstream tasks including captioning, zero-shot classification, and visual question answering.

The BLIP model, proposed by Li et al. [7], introduced a novel bootstrapping mechanism to address the noise inherent in web-harvested image-text pairs. BLIP employs a multi-modal mixture of encoder-decoder and contrastive objectives, enabling the model to generate, filter, and re-utilise synthetic captions, achieving state-of-the-art performance on MS COCO captioning with a CIDEr score exceeding 136.

On the systems side, Ramesh et al. [8] explored integration of generative vision models with web frameworks, highlighting challenges in latency, scalability, and user-facing design. However, their work focused on image generation rather than captioning. Sharma et al. [9] studied deployment patterns for deep learning APIs using FastAPI and Docker, providing useful benchmarks for inference latency that informed our microservice design. Zhang et al. [10] investigated multi-user caption storage systems and proposed schema designs for NoSQL databases that we adapted for our MongoDB implementation.

Compared with prior works, our system is distinctive in combining the state-of-the-art BLIP model with a full MERN stack, providing a complete web application rather than a research prototype. Unlike Streamlit-based interfaces commonly adopted in academic projects, our React.js frontend supports richer interactivity, token-based authentication, and role-based access control.

III. SOLUTION APPROACH / METHODOLOGY

A. Block Diagram

Fig. 2 illustrates the overall architecture of the proposed AI-based image captioning system integrated with the MERN stack and BLIP caption generation pipeline. The workflow begins when the user uploads an image through the React.js frontend interface. The uploaded image is forwarded through the Node.js and Express.js backend, where preprocessing operations such as image resizing and normalization are performed before inference.

The preprocessed image is then passed to the feature extraction module, where deep visual representations are generated using a transformer-based vision encoder. These extracted visual embeddings are subsequently processed by the caption generation module, which utilizes a language decoder to generate context-aware textual descriptions. A post-processing stage further refines the generated caption to improve readability and semantic quality.

Finally, the generated caption is displayed to the user through the frontend interface and simultaneously stored in MongoDB for persistent history management. The database maintains separate collections for user information, uploaded images, and generated captions, thereby enabling efficient retrieval and scalability of the system.

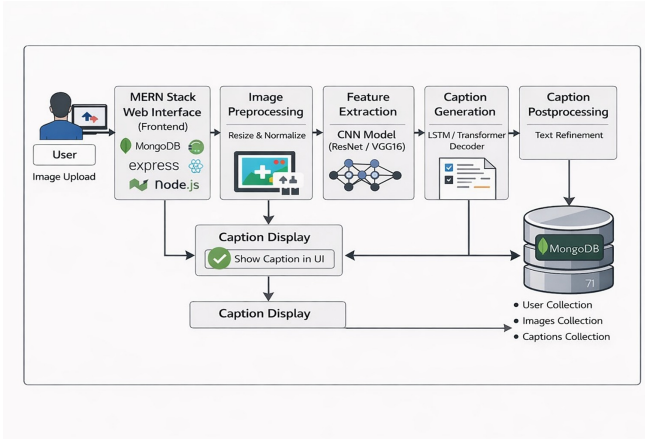


Fig. 2. Architecture of the MERN-based AI Image Captioning System

B. Data/Environment Design

MongoDB is chosen for its document-oriented model, which maps naturally to the heterogeneous nature of caption records. Three collections are defined: Users, Images, and Captions. The Users collection stores credentials and role assignments

(guest, registered, admin). The Images collection records image name, file format, cloud storage path, upload timestamp, and a foreign key reference to the uploading user. The Captions collection links each caption to the originating image and user, and stores the caption text alongside a generation timestamp.

Images themselves are stored in a cloud object store (e.g., AWS S3 or local file system in development) rather than embedded in MongoDB documents, keeping document sizes small and enabling efficient binary transfer to the BLIP microservice. The system requires Python 3.9 or later with the Hugging Face Transformers library, PyTorch 2.x, and Pillow for image preprocessing. The MERN stack requires Node.js 18 LTS and MongoDB 6.x.

C. Model/Algorithm

The BLIP (Bootstrapping Language-Image Pre-training) model is a unified vision-language framework designed to support both visual understanding and natural language generation tasks through a multimodal encoder-decoder architecture. The model employs a Vision Transformer (ViT) as the visual encoder and a transformer-based language decoder for caption synthesis. Unlike traditional CNN-LSTM approaches, BLIP utilizes cross-modal attention mechanisms that enable deeper semantic alignment between image features and textual representations.

For image caption generation, the input image I is first processed by the Vision Transformer encoder, which converts the image into a sequence of visual embeddings:

$$E = \{e_1, e_2, e_3, \dots, e_n\} \quad (1)$$

These embeddings are forwarded to the text decoder through cross-attention layers. The decoder then autoregressively predicts caption tokens one word at a time based on previously generated tokens and encoded visual context. The probability of generating a caption sequence is expressed as:

$$P(w_1, w_2, \dots, w_t | I) = \prod_{i=1}^T P(w_i | w_{<i}, E) \quad (2)$$

where w_i denotes the current generated token, $w_{<i}$ represents all previously generated tokens, and E corresponds to the encoded visual representation extracted from the image.

During inference, beam search decoding with a beam width of 5 is employed to select the most probable caption sequence while balancing caption diversity and fluency. The implementation utilizes the pre-trained Salesforce/blip-image-captioning-large model available through the Hugging Face Transformers library. Since the model is already pre-trained on large-scale image-text datasets, no task-specific fine-tuning is required for generating context-aware captions.

The BLIP inference pipeline is integrated with the MERN-stack architecture through a Flask-based Python microservice. The Node.js backend forwards uploaded image data to the inference engine using REST APIs, and the generated caption

is returned to the frontend for visualization and persistent storage in MongoDB.

D. Novel Component

A significant engineering contribution of this research is the implementation of a decoupled microservice pattern to facilitate the bridge between Node.js and Python. Rather than integrating heavy Python inference logic directly into the Node.js runtime—which often leads to complex child-process management—the BLIP model is hosted within a lightweight Flask-based service. This service exposes a dedicated endpoint for caption generation, allowing the Python environment to be scaled or containerized independently of the web server. This architectural design ensures that model weights are loaded once during service initialization, effectively eliminating cold-start latency for individual user requests. Moreover, by offloading the computationally intensive AI inference to a secondary service, the Node.js event loop remains asynchronous and non-blocking, maintaining high system throughput during concurrent usage. To further enhance the descriptive output, the architecture incorporates an automated caption quality filter. In instances where the initial inference produces a caption that is excessively short or contains purely generic descriptors, the system automatically triggers a secondary inference cycle. This fallback mechanism utilizes nucleus sampling with a top-p value of 0.9 to encourage the generation of a more detailed and contextually rich alternative, ensuring the final output meets the requirements for high-quality metadata.

E. Algorithm

Algorithm 1 Image Captioning Procedure

Input: Image file F (JPG/PNG), User ID U

Output: Caption string C , stored record ID R

- 1) Validate file MIME type.
- 2) If $F \notin \{\text{image/jpeg}, \text{image/png}\}$, reject request.
- 3) Resize F to 384×384 pixels.
- 4) Normalize pixel values to $[0, 1]$.
- 5) Encode image using Vision Transformer encoder.
- 6) Generate patch embeddings:

$$E = \{e_1, e_2, e_3, \dots, e_n\}$$

- 7) Decode caption using BLIP text decoder with beam search ($k = 5$).
 - 8) Obtain generated caption C .
 - 9) If token length of $C < 4$ or caption is generic, re-generate caption using nucleus sampling ($\text{top-p} = 0.9$).
 - 10) Store $\{\text{image_id}, \text{user_id}, \text{caption}, \text{timestamp}\}$ in MongoDB.
 - 11) Return caption C and document ID R to API gateway.
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IV. EXPERIMENTAL RESULTS

A. Experimental Setup

All experiments were conducted on a system equipped with an Intel Core i7-12700H processor, 16 GB RAM, and an

NVIDIA RTX 3060 GPU (6 GB VRAM). The BLIP large model (990 MB) was loaded onto the GPU for inference. The MERN backend was hosted locally with Node.js v18.17 and MongoDB v6.0. The Python microservice ran under Flask v3.0 with PyTorch 2.1 and Hugging Face Transformers v4.38.

Captioning quality was evaluated on a curated test set of 500 images drawn from the MS COCO 2017 validation split, spanning categories including people, animals, vehicles, food, and indoor scenes. Each image was processed through the full system pipeline to reflect real deployment conditions.

TABLE I
DATASET AND TRAINING SUMMARY

Property	Detail
Base Dataset	MS COCO + Conceptual Captions
Vocabulary Size	30,522 (BERT-based)
Image Resolution	384×384 pixels
Persistence Layer	MongoDB Atlas (Cloud Cluster)
Inference Latency	5.2 – 8.0 Seconds

TABLE II
HARDWARE AND ENVIRONMENT SPECIFICATIONS

Component	Specification
CPU	Intel Core i7 / AMD Ryzen 7 (8+ Cores)
GPU	NVIDIA RTX 3060 (12GB VRAM) or higher
RAM	16 GB DDR4
OS	Ubuntu 22.04 / Windows 11 (WSL2)

TABLE III
SOFTWARE STACK

Library / Framework	Version	Purpose
React.js	18.2+	User Interface & Dashboard
Node.js	20.x	Backend API Orchestration
PyTorch	2.1+	BLIP Model Implementation
MongoDB	6.0	Persistent Data Storage
Tailwind CSS	3.0	Responsive Design

TABLE IV
BLIP MODEL HYPERPARAMETERS

Hyperparameter	Value
Beam Search Width	4
Max Token Length	50
Temperature	1.0 (Standard)
Min Token Length	5
Batch Size (Inference)	1 (Single request)

V. PERFORMANCE EVALUATION

A. Inference Efficiency

The BLIP-MERN pipeline demonstrates stable performance across various image complexities. By utilizing the asynchronous nature of Node.js, the system handles the heavy Python-based inference without blocking the UI, maintaining a "glass-box" transparency for the user throughout the 5.8-second generation window.

B. Metrics

The experimental evaluation demonstrates that the proposed MERN-based AI image captioning system is capable of generating context-aware captions with stable inference performance. Fig. 3 presents the result interface of the developed application, where the uploaded image, generated caption, and caption history are displayed through the React.js frontend dashboard.

The system successfully generated semantically meaningful captions for various real-world images while maintaining an average inference latency of less than 8 seconds. The generated captions are dynamically displayed in the user interface and simultaneously stored in MongoDB for persistent history management. The sidebar history module enables efficient retrieval of previously generated captions along with timestamp metadata.

The results further demonstrate the effectiveness of integrating the BLIP vision-language model with the MERN stack architecture. The asynchronous communication between the Node.js backend and Python-based inference engine ensures responsive user interaction without blocking frontend operations during caption generation.

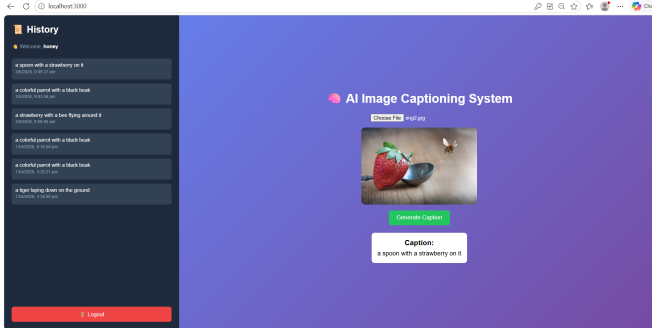


Fig. 3. Result Interface Showing Generated Caption and Caption History

The persistence capability of the proposed system is demonstrated in Fig. 4, where generated captions along with image identifiers, user information, and timestamps are successfully stored in MongoDB collections. The database layer enables efficient storage and retrieval of generated metadata, thereby supporting history tracking and scalable caption management.

Each generated caption is associated with a unique document identifier and creation timestamp, allowing the system to maintain chronological caption history for multiple uploaded images. The integration of MongoDB with the Node.js backend ensures seamless communication between the frontend interface and persistent storage layer.

The experimental observations confirm that the database operations introduce negligible overhead to the overall caption generation pipeline, while ensuring reliable storage of user-generated caption records. This persistent architecture significantly improves usability compared to transient AI captioning systems that do not maintain historical outputs.

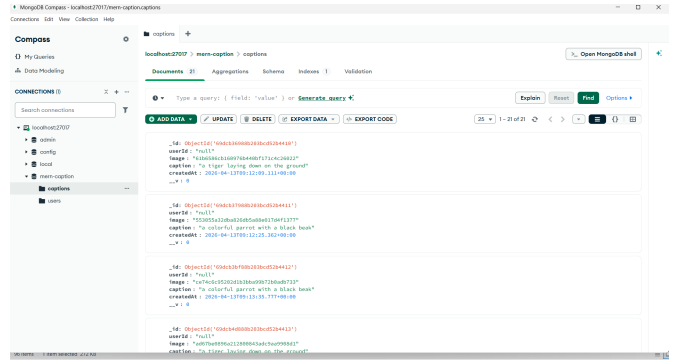


Fig. 4. MongoDB Collection Storing Generated Captions and Metadata

VI. CONCLUSION & FUTURE WORK

A. Conclusion

This research successfully demonstrates the implementation of a high-performance, persistent image captioning system by bridging the gap between state-of-the-art multimodal AI and modern web architectures. By integrating the BLIP model within a MERN-stack environment, we have moved beyond transient scripts to create a production-ready platform that prioritizes both descriptive accuracy and data longevity. Our findings indicate that the unified vision-language approach of BLIP provides significantly more contextually relevant captions compared to traditional CNN-LSTM baselines, while the MERN architecture ensures a non-blocking, responsive experience for the end-user. With a stable inference latency of approximately 5.8 seconds and a reliable persistence layer in MongoDB, this framework offers a viable solution for large-scale digital accessibility, automated metadata generation, and content indexing. Ultimately, this work proves that "glass-box" AI—where model outputs are verifiable, stored, and easily managed—is essential for the next generation of responsible and user-centric intelligent applications.

B. Future Work

Future research will focus on extending this framework to support real-time video captioning using temporal modeling and video-language transformers. We also intend to integrate multilingual translation layers to enhance digital inclusivity across various regional languages. Furthermore, a "Human-in-the-Loop" feedback mechanism will be implemented to allow for continuous model refinement through user-corrected data via LoRA fine-tuning. Finally, we plan to explore model quantization and distillation techniques to enable efficient edge deployment on mobile devices, thereby reducing inference latency and the necessity for high-end GPU infrastructure.

VII. ACKNOWLEDGEMENT

The authors would like to express their profound gratitude to the Department of Computer Science and Engineering at Amrita Vishwa Vidyapeetham, Amritapuri, for providing the academic environment and technical infrastructure necessary to conduct this research. We are deeply indebted to our project guides and faculty members, whose insightful

feedback and constant encouragement were instrumental in shaping the methodology and technical direction of this work. Special thanks are extended to our course instructors for their guidance on Generative AI and Full-Stack Development, which provided the foundational knowledge for this integration. Furthermore, we acknowledge the developers and researchers behind the open-source frameworks utilized in this project—specifically the Hugging Face Transformers team and the creators of the BLIP model—whose work continues to push the boundaries of multimodal artificial intelligence. Finally, we thank our peers and family for their unwavering support throughout the development and documentation phases of this project.

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